

# Improved Delayed Neutron Group Parameters for Molten Salt Reactors

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# Outline

## 1 Introduction

Defining Terms  
Motivation  
Objectives

## 2 Background

DNP Groups  
MSR Phenomena

## 3 Methodology

MoSDeN Overview  
0D Scaled Model

## 4 Preliminary Results

Analysis of Yields and Counts  
Sensitivity Studies  
Verification

## 5 Proposed Work

0D Flow Model  
Validation  
Transient Analyses  
Summary

## Delayed neutron precursors (DNPs) emit delayed neutrons

Delayed neutron precursors (DNPs) are nuclides that undergo  $\beta^-$  decay, producing a daughter, called an emitter, which emits a delayed neutron [22].

Relevant data for DNPs includes:

$P_{n,i}$  = the emission probability [#]

$\tau_i$  = the half-life of DNP  $i$  [s]

$\chi_{d,i}$  = the delayed neutron energy spectrum from DNP  $i$  [-]

$N_i$  = the number of atoms of  $i$  [#].

**Accurately modeling the DNPs is important for transient simulations, which are important for safety, security, and safeguards.**

## DNP groups approximate DNPs

A small number of artificial DNPs, known as DNP groups, are used to approximate the DNPs to reduce computational cost.

The three parameters for DNP groups are:

$\bar{\nu}_{d,k}$  = the delayed neutron yield of DNP group  $k$  [-]

$\tau_k$  = the half-life of DNP group  $k$  [s]

$\chi_{d,k}$  = the delayed neutron energy spectrum of DNP group  $k$  [-]

**DNP groups are commonly used to approximate the DNPs in a light water reactor [3].**

## Moltres provides multiphysics MSR simulation capability



Moltres is a multiphysics reactor simulation tool built on the MOOSE framework [19].

### Moltres enables:

- Multigroup neutron diffusion
- DNP recirculation
- Heat transfer and fluid flow

**Moltres will be used for the proposed safety, security, and safeguards transient simulations.**



## Safety, security, and safeguards are important

Safety, security, and safeguards by design is the act of developing a facility with features and processes that inherently address safety, security, and safeguards requirements from the earliest stages of design.

- *Safety* - Protect the public and minimize accident risk [14].
- *Security* - Prevent actors from acquiring information, stealing nuclear material, or committing sabotage [32].
- *Safeguards* - Impede the diversion or undeclared production of nuclear material for weapons by a host state [15].

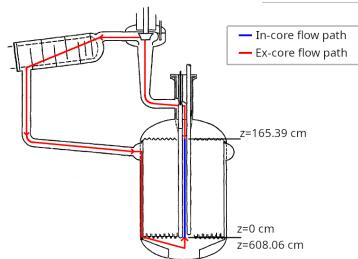
**Safety, security, and safeguards by design requires accurate transient simulations.**

# MSR phenomena are not captured in conventional parameters

## MSR phenomena that affect DNP group parameters:

- Recirculation of fuel salt
- Elemental chemical removal from reprocessing

**Conventional DNP group parameters do not account for MSR recirculation or chemical removal.**



**Figure 1:** The flow path and bounds for the in-core region of the MSRE, overlayed on the MSRE loop diagram from [10].



## This work will capture MSR phenomena in the DNP groups

**The primary objective of this work is to generate improved DNP group parameters for MSRs.**

The objectives of this work are to:

-  Create a Python package (Molten Salt Delayed Neutron (MoSDeN)) for DNP group parameter generation
-  Verify and validate the improved DNP group parameters
-  Evaluate the impact of these improved parameters on transient simulations for safety, security, and safeguards problems



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## There are two approaches to calculating DNP group parameters

DNP group parameter calculation follows the following steps:

- ① Irradiate a fissile sample
- ② Measure/calculate a delayed neutron count rate
- ③ Determine the DNP group parameters that best fit that count rate

**The specific steps depend on which approach is used.**

### Macroscopic approach

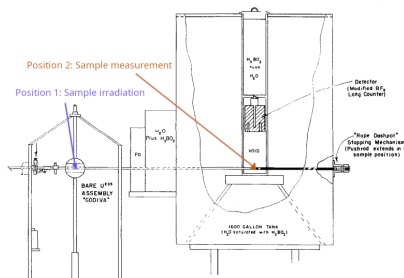
The delayed neutron count rate is *measured*.

### Microscopic approach

The delayed neutron count rate is *calculated*.

## The macroscopic approach irradiates a sample physically

The delayed neutron count rate ( $\dot{n}_d$ ) is directly measured after sample irradiation.



**Figure 2:** Simplified experimental setup for measuring delayed neutron emission from irradiation of a 3 gram sample in Godiva referenced from [18].



## The microscopic approach irradiates a sample computationally

**The delayed neutron count rate is calculated from aggregated experimental data.**

$$\dot{n}_d(t) = \sum_i^I P_{n,i} \lambda_i N_i(t), \quad (1)$$

where

$I$  = the number of DNPs [#]

$\dot{n}_d$  = the delayed neutron count rate [ $\# \cdot s^{-1}$ ]

$P_{n,i}$  = the average number of delayed neutrons emitted from decay of nuclide  $i$ , or the emission probability [#]

$\lambda_i$  = the decay constant of DNP  $i$  [ $s^{-1}$ ]

$N_i(t)$  = the number of atoms of DNP  $i$  [#].



## There is a general equation for sample irradiation

The  $I$  DNPs are approximated by  $K$  groups.

### Intermediate Irradiation

$$\dot{n}_d(T, t) \approx \dot{F}_s \sum_{k=1}^K \bar{\nu}_{d,k} e^{-\lambda_k t} (1 - e^{-\lambda_k T}) \quad (2)$$

$\dot{F}_s$  = The fission rate in the sample [ $\# \cdot s^{-1}$ ]

$F_s$  = The number of fissions in the sample [ $\#$ ]

$T$  = The time over which the sample is irradiated [ $s$ ]



There are two common irradiation types used to generate groups

### Intermediate Irradiation

$$\dot{n}_d(T, t) \approx \dot{F}_s \sum_{k=1}^K \bar{\nu}_{d,k} e^{-\lambda_k t} (1 - e^{-\lambda_k T}) \quad (3)$$

### Pulse irradiation ( $T \rightarrow 0$ )

$$\dot{n}_d(t) \approx F_s \sum_{k=1}^K \bar{\nu}_{d,k} \lambda_k e^{-\lambda_k t} \quad (4)$$

### Saturation irradiation ( $T \rightarrow \infty$ )

$$\dot{n}_d(t) \approx \dot{F}_s \sum_{k=1}^K \bar{\nu}_{d,k} e^{-\lambda_k t} \quad (5)$$

The DNP parameters  $\bar{\nu}_{d,k}$  and  $\lambda_k$  make  $\dot{n}_d(t)$  match using exponential peeling [5, 17], least squares [4, 18], or half-lives of specific DNPs [28].

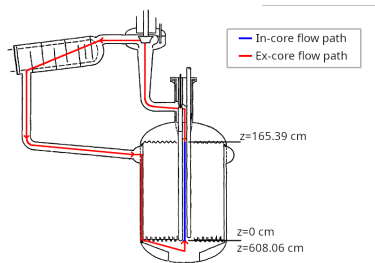
## Fuel salt recirculation affects concentration of DNPs

The fission normalized yields vary due to differences in DNP concentrations induced by flowing fuel.

In-core and ex-core salt recirculation can be modeled using:

- **Scaled flux method** [2]
- **Piecewise 0D method**
- **Spatially resolved method** [31, 29]

**For a recirculating fissile sample, the piecewise 0D model offers a good balance of accuracy and speed.**



**Figure 3:** The flow path and bounds for the in-core region of the MSRE, overlaid on the MSRE loop diagram from [10].



## Chemical reprocessing affects DNPs concentrations

### Online chemical reprocessing includes:

- Fission product removal
- Fresh fuel addition

Conventional DNP group parameters do not account for these effects.

**For example, a reprocessing scheme with 100% of all DNPs chemically removed should not have the same DNP groups.**



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  - MoSDeN Overview
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## MoSDeN generates improved DNP group parameters

MoSDeN is an open-source Python package in development that will enable users to generate improved DNP group parameters.

### MoSDeN currently enables:

- Use of various data inputs for half-lives ( $\tau$ ), emission probabilities ( $P_n$ ), and cumulative fission yields (CFY)
  - ENDF-6 formatted data
  - OpenMC chain data
  - Custom data
- Calculation of concentrations and delayed neutron count rates of an irradiated sample
- Calculation of the DNP group parameters that fit the calculated count rate using non-linear least squares

**MoSDeN currently offers the 0D scaled model, which only enables saturation irradiations to be modeled.**



## The saturation irradiation is a non-linear least squares problem

The group parameters are calculated by equating the delayed neutron count rates.

$$\dot{F}_s \sum_{k=1}^K \bar{\nu}_{d,k} e^{-\lambda_k t} \approx \sum_i^I P_{n,i} \lambda_i N_i(t) \quad (6)$$

$$\dot{F}_s \begin{bmatrix} e^{-\lambda_1 t_0} & \dots & e^{-\lambda_K t_0} \\ \vdots & \ddots & \vdots \\ e^{-\lambda_1 t_m} & \dots & e^{-\lambda_K t_m} \end{bmatrix} \begin{bmatrix} \bar{\nu}_{d,1} \\ \vdots \\ \bar{\nu}_{d,K} \end{bmatrix} \approx \begin{bmatrix} \dot{n}_d(t_0) \\ \vdots \\ \dot{n}_d(t_m) \end{bmatrix} \quad (7)$$

$$A \vec{x} \approx \vec{b} \quad (8)$$



## The residual best fits the delayed neutron count rate

$$r = \min_x \left\| \frac{\vec{b} - A\vec{x}}{\vec{b}} \right\|_2^2, \quad (9)$$

where

$$\begin{aligned} \tau_k &\in [1 \text{ ms}, 1000 \text{ s}] \\ \vec{v}_{d,k} &\in [0, 1]. \end{aligned}$$

**MoSDeN uses the SciPy package's optimize module to solve this non-linear least squares problem [34].**



## Uncertainties are tracked using Monte Carlo sampling

The process for uncertainty tracking is as follows:

- 1 Normally sample  $P_{n,i}$ ,  $\tau_i$ , and  $N_i(t)$  within their uncertainties for every DNP
- 2 Calculate the DNP group parameters using non-linear least squares
- 3 Record the results and repeat

After all simulations finish:

- 1 Groups are defined based on half-lives (e.g. longest lived half-life and yield are group one)
- 2 The mean values and standard deviations are calculated



## MoSDeN uses a newly-derived form of the group fit for MSRs

Assuming the sample experiences a square-wave power history during in-core and ex-core times, an MSR specific group fit equation can be derived.

$$\frac{F(\tau)}{F_0} = \sum_{j=0}^J H(\tau - j \cdot \tau_{net}) - H(\tau - j \cdot \tau_{net} - \tau_{in}) \quad (10)$$

where

$F(\tau)$  = the fission rate history  $[\frac{\#}{s}]$ .

$H$  = the Heaviside function  $[-]$

$$\tau_{net} = \tau_{in} + \tau_{ex}$$

= the total time per circulation  $[s]$

$$J = \left\lfloor \frac{T}{\tau_{net}} \right\rfloor$$

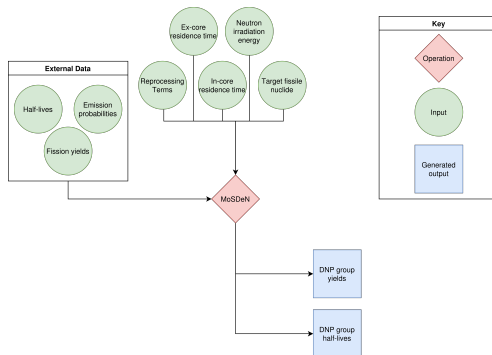
= the number of recirculations  $[\#]$

$F_0$  = the magnitude of the constant fission rate  $[\frac{\#}{s}]$ .



## The 0D scaled model is implemented in MoSDeN

**MoSDeN uses simulation parameters and external data to generate improved DNP parameters.**



**Figure 4:** Flow diagram of 0D scaled model used to generate improved DNP group parameters.



## The 0D scaled model is fast but uses many approximations

The equilibrium concentration of each DNP is calculated for the 0D scaled model as [13]:

$$N_i = \frac{f_{in} CFY_i}{\lambda_i}, \quad (11)$$

where

$f_{in}$  = the fraction of time fuel-salt spends in the core [-]

$CFY_i$  = the cumulative fission yield of nuclide  $i$  [-].

The limitations of the 0D model include:

- Concentration calculation does not include cross sections
- The concentration is only available at equilibrium
- The depletion chain is not tracked, so chemical removal is underestimated



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- ② Background
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  - MoSDeN Overview
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  - Verification
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  - Summary

## The parameters for the 0D scaled model analyses

**Table 1:** Default parameters for 0D scaled model simulations.

| Parameter               | Value            | Units |
|-------------------------|------------------|-------|
| Fissile Nuclide         | $^{235}\text{U}$ | [-]   |
| Incident Neutron Energy | 0.0253           | [eV]  |
| Fission Yields          | ENDF/B-VII.1 [6] | [-]   |
| Half-lives              | IAEA [9]         | [-]   |
| Emission Probability    | IAEA [9]         | [-]   |
| Final Decay Time        | 1200             | [s]   |
| Decay Time Spacing      | Logarithmic      | [-]   |
| Decay Time Nodes        | 300              | [-]   |
| Sampled Simulations     | 1000             | [-]   |
| $\tau_{in}$             | 9 [11]           | [s]   |
| $\tau_{ex}$             | 16 [11]          | [s]   |

# The chemical removal rates for the 0D scaled model analyses

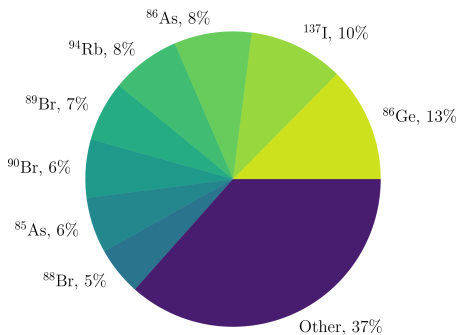
Table 2: MSBR Online Reprocessing Removal Rates from [24].

| Reprocessing Group | Element(s)                             | Removal Rate [ $s^{-1}$ ] |
|--------------------|----------------------------------------|---------------------------|
| Rare Earths        | Y, La, Ce, Pr, Nd, Pm, Sm, Gd          | $2.32 \times 10^{-7}$     |
| Rare Earths        | Eu                                     | $2.32 \times 10^{-8}$     |
| Noble Metals       | Se, Nb, Mo, Tc, Ru, Rh, Pd, Ag, Sb, Te | $5.00 \times 10^{-2}$     |
| Seminoble Metals   | Zr, Cd, In, Sn                         | $5.79 \times 10^{-8}$     |
| Gases              | Kr, Xe                                 | $5.00 \times 10^{-2}$     |
| Volatile Fluorides | Br, I                                  | $1.93 \times 10^{-7}$     |
| Discard            | Rb, Sr, Cs, Ba                         | $3.37 \times 10^{-9}$     |
| Salt Discard       | Th, Li, Be, F                          | $3.37 \times 10^{-9}$     |
| Protactinium       | Pa                                     | $3.86 \times 10^{-6}$     |
| Higher Nuclides    | Np, Pu                                 | $1.98 \times 10^{-9}$     |

## The total delayed neutron yield is dominated by a few DNPs

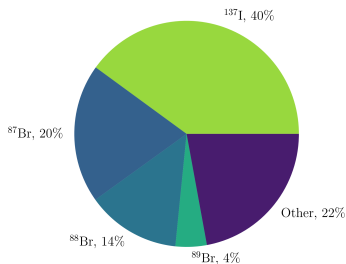
The delayed neutron yield from each DNP is given as [13, 20]:

$$\bar{\nu}_{d,i} = P_{n,i} C F Y_i. \quad (12)$$

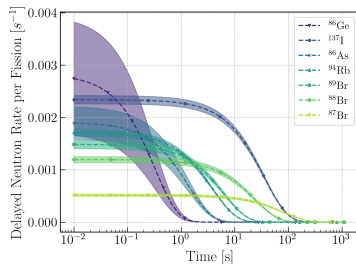


**Figure 5:** Relative delayed neutron yield from each DNP after a thermal saturation irradiation of <sup>235</sup>U using the 0D scaled model.

## The delayed neutron count rate is dominated by a few DNPs



**Figure 6:** Relative delayed neutron counts from each DNP after a thermal saturation irradiation of  $^{235}\text{U}$  using the 0D scaled model.



**Figure 7:** Relative delayed neutron count rates after a thermal saturation irradiation of  $^{235}\text{U}$  using the 0D scaled model.



## Chemical removal and simulation parameters affect the results

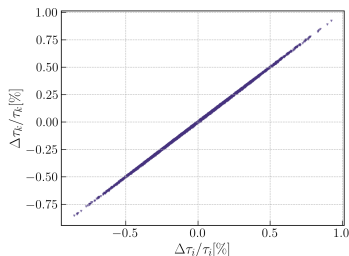
### Sensitivity studies of the chemical removal show:

- Only the noble metal and gas reprocessing groups have a non-negligible impact on DNP group parameters
- Chemical removal (assuming 50% in-core fraction) in the 0D scaled model decreases the total delayed neutron yield by 12 pcm

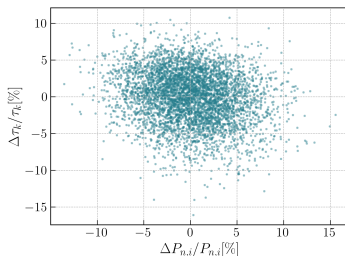
### Sensitivity studies of the simulation parameters show:

- Logarithmic time node spacing improves the DNP group parameter fit compared to linear spacing with the same number of nodes
- Increasing the number of time nodes improves the results (up to  $\approx 800$ )
- Increasing the length of time for delayed neutron count data collection improves results (up to  $\approx 40$  minutes)

## PCCs convey sensitivity of DNP group parameters to data



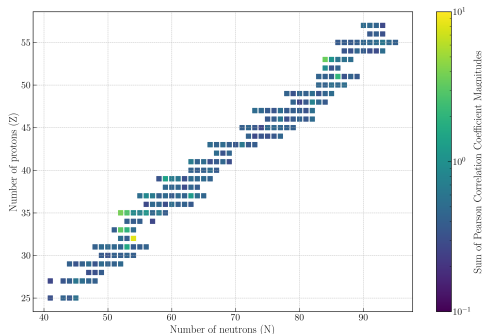
**Figure 8:** DNP group 1 half-life variation due to  $^{87}\text{Br}$  half-life variation with a PCC of 1.00.



**Figure 9:** DNP group 3 half-life variation due to  $^{87}\text{Br}$  emission probability variation with a PCC of -0.19.

The half-life of  $^{87}\text{Br}$  strongly correlates to the group one half-life with a Pearson Correlation Coefficient (PCC) magnitude close to 1.0, while its emission probability does not strongly correlate to the group three half-life.

# The PCC magnitudes show the most sensitive DNPs



**Figure 10:** The summed PCC magnitudes after a thermal saturation irradiation of  $^{235}\text{U}$  using the 0D scaled model.

**Table 3:** The largest summed PCC magnitudes after a thermal saturation irradiation of  $^{235}\text{U}$  using the 0D scaled model.

| Nuclide           | $PCC_i$ |
|-------------------|---------|
| $^{86}\text{Ge}$  | 8.06    |
| $^{87}\text{Br}$  | 3.82    |
| $^{137}\text{I}$  | 3.52    |
| $^{85}\text{As}$  | 3.14    |
| $^{88}\text{Br}$  | 2.77    |
| $^{137}\text{Sb}$ | 2.08    |
| $^{86}\text{As}$  | 1.97    |
| $^{84}\text{Ga}$  | 1.62    |
| $^{98}\text{Y}$   | 1.42    |
| $^{100}\text{Rb}$ | 1.22    |

# The uncertainty-weighted PCC shows critical data needs

## Uncertainty-weighted PCC $U_i$ :

$$U_i = \sum_v \frac{\sigma_{i,v}}{x_{i,v}} \sum_g |PCC_{i,v,g}|, \quad (13)$$

where

$U_{i,v}$  = the uncertainty-weighted sensitivity of  
DNP value  $v$  for DNP  $i$  [-]

$\sigma_{i,v}$  = the uncertainty in the DNP value  $v$   
for DNP  $i$  [-]

$x_{i,v}$  = the DNP value  $v$ , representing the half-life,  
emission probability, or concentration of DNP  $i$  [-]

$PCC_{i,v,g}$  = the PCC for DNP  $i$  and value  $v$  with  
DNP group parameter  $g$  [-].

**Table 4:** The largest summed uncertainty-weighted PCC magnitudes after a thermal saturation irradiation of  $^{235}\text{U}$  using the 0D scaled model.

| Nuclide           | $U_i$ |
|-------------------|-------|
| $^{86}\text{Ge}$  | 1.71  |
| $^{124}\text{Ag}$ | 0.41  |
| $^{120}\text{Ag}$ | 0.36  |
| $^{114}\text{Tc}$ | 0.36  |
| $^{147}\text{Xe}$ | 0.34  |
| $^{104}\text{Nb}$ | 0.29  |
| $^{111}\text{Mo}$ | 0.27  |
| $^{122}\text{Pd}$ | 0.27  |
| $^{98}\text{Y}$   | 0.26  |
| $^{103}\text{Zr}$ | 0.26  |

## The DNP group parameters accurately represent the DNP data

The total delayed neutron yield ( $\bar{\nu}_d$ ) from the DNP group parameters should align with the DNP data:

$$\sum_{k=1}^K \bar{\nu}_{d,k} \approx \sum_{i=1}^I P_{n,i} CFY_i = \bar{\nu}_d \quad (14)$$

The  $K$  groups give  $\bar{\nu}_d = 2.24 \pm 0.13$  per 100 fissions, while the  $I$  DNPs give  $\bar{\nu}_d = 2.25 \pm 0.15$ .

The average half-life ( $\bar{\tau}$ ) from the DNP group parameters should align with the DNP data:

$$\ln(2) \sum_{k=1}^K \frac{\bar{\nu}_{d,k}}{\lambda_k} \approx \ln(2) \sum_{i=1}^I \frac{P_{n,i} CFY_i}{\lambda_i} = \bar{\tau} \quad (15)$$

The  $K$  groups give  $\bar{\tau} = 6.4 \pm 0.3s$ , while the  $I$  DNPs give  $\bar{\tau} = 6.4 \pm 0.4s$ .

**The total delayed neutron yield and average half-life agree between the model and the data.**



## The DNP group parameters do not align with the literature

The total delayed neutron yield ( $\bar{\nu}_d$ ) from the DNP group parameters should align with the literature [22, 27, 6, 25, 21, 3, 7, 18, 8, 26, 30, 33, 35]:

The  $K$  groups give  $\bar{\nu}_d = 2.24 \pm 0.13$  per 100 fissions, while the literature gives  $1.48 \leq \bar{\nu}_d \leq 1.91$ .

The average half-life ( $\bar{\tau}$ ) from the DNP group parameters should align with the literature [25, 21, 3, 18, 8, 26, 30, 33, 35]:

The  $K$  groups give  $\bar{\tau} = 6.4 \pm 0.3s$ , while the literature gives  $7.7 \leq \bar{\tau} \leq 9.5s$ .

**This indicates an issue with short-lived DNPs, likely from the selected nuclear data and the 0D model assumptions.**

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  - Sensitivity Studies
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The objectives of this work are to:

-  Create a Python package (MoSDeN) for DNP group parameter generation
-  Verify and validate the improved DNP group parameters
-  Evaluate the impact of these improved parameters on transient simulations for safety, security, and safeguards problems

## The proposed work consists of three categories

### Development of the 0D flow model:

- Enables improved DNP group parameters to account for residence time
- Includes development of the MoSDeN Python package
- Includes various other analyses and developments

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- Enables comparison between the 0D flow and 0D scaled models

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### Analysis of safety, security, and safeguards using MSRE transients:

- Demonstrates the effects of the improved DNP group parameters
- Investigate how conventional DNP group parameters may predict differing behavior than the improved parameters

## The 0D flow model will introduce new features and analyses

### New features in the 0D flow model:

- Full depletion chain modeling in OpenMC
  - Improving chemical removal modeling
  - Enabling residence time analysis
  - Enabling non-equilibrium DNP concentrations
  - Improving DNP concentration accuracy
- Calculation of the improved DNP group spectra ( $\chi_{d,k}$ )

### New analyses using the 0D flow model:

- Novel analysis for residual minimization using pulse and saturation irradiation data [4, 18]
- Comparison of various nuclear datasets [13, 22, 20]

## The MSRE will be used for validation

### Necessary components will be computed/generated:

- Improved DNP group parameters for the MSRE
- Multi-group cross sections using OpenMC for the MSRE
- Upper and lower plena added to an existing MSRE Moltres model

### Quantities of interest will be computed and compared with experimental data for validation:

- Calculation of  $\beta_{eff}$  using the prompt method for stationary and flowing fuel salt using improved and conventional DNP group parameters [12]
- Simulate pump start-up and coast-down transients to compare reactivity responses to experimental data [23]

## Safety-related transients may disrupt safe reactor operation

Safety-related transients that will be modeled are:

- Unprotected loss of heat sink (ULOHS)
- Unprotected loss of flow (ULOF)
- Unprotected transient overpower (UTOP)
- Fuel salt overcooling (FSOC)

Table 5: Implementation of transients using temperature, reactivity, and velocity changes

| Transient | $T$                     | $\rho$               | $v$                |
|-----------|-------------------------|----------------------|--------------------|
| ULOHS     | $T_{in} = T_{out}$      | -                    | -                  |
| ULOF      | -                       | -                    | $v \approx e^{-t}$ |
| UTOP      | -                       | $\rho_{max} = 0.1\%$ | -                  |
| FSOC      | $T_{in} = 0.9 T_{in,0}$ | -                    | -                  |

## Sabotage may be used as a diversion

This framework assumes a hostile actor and an insider with the objective to disrupt operations [32].

Sabotage-initiated transients that will be modeled are:

- All safety-related transients as single events
- Combined UTOP and ULOHS
- Combined UTOP and ULOF
- Combined UTOP and FSOC

Table 6: Implementation of transients using temperature, reactivity, and velocity changes

| Transient | $T$                    | $\rho$               | $v$                |
|-----------|------------------------|----------------------|--------------------|
| ULOHS     | $T_{in} = T_{out}$     | -                    | -                  |
| ULOF      | -                      | -                    | $v \approx e^{-t}$ |
| UTOP      | -                      | $\rho_{max} = 0.1\$$ | -                  |
| FSOC      | $T_{in} = 0.9T_{in,0}$ | -                    | -                  |

## SNM diversion may be detected using reactor kinetics

Special nuclear material (SNM) diversion scenarios considered are [32, 16, 1]:

- One significant quantity (SQ) of  $^{235}\text{U}$  (75 kg) after one year in the MSRE
  - Removing  $^{235}\text{U}$  will decrease the delayed neutron fraction
- One SQ of Pu (8 kg) after one year in the MSRE
  - Removing Pu will increase the delayed neutron fraction

The DNP group parameters will then be used in a negative reactivity step insertion, and the resulting power responses will be compared [36].

Each scenario will have four sets of DNP group parameters to compare:

- Conventional parameters with/without diversion
- Improved parameters with/without diversion

**The change in fuel composition and the MSR phenomena affect the ability to detect SNM diversion.**

# Summary

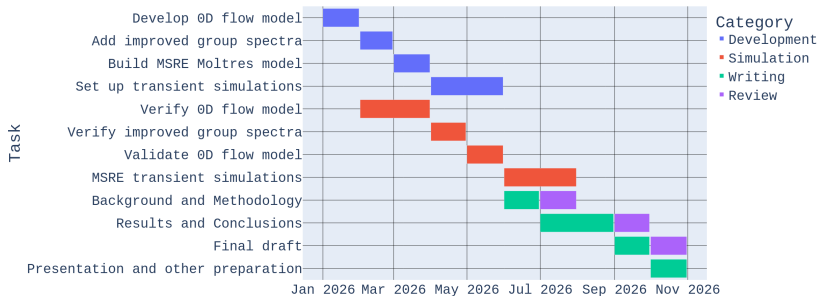


Figure 11: A Gantt chart of the proposed work.

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## MoSDeN uses a newly-derived form of the group fit for MSRs



Assuming the sample experiences a square-wave power history during in-core and ex-core times, an MSR specific group fit equation can be derived.

$$\begin{aligned}
 \frac{\dot{n}_{d,k}(T, t)}{F_0 \bar{\nu}_{d,k}} = & \overbrace{e^{-\lambda_k t} \sum_{j=0}^{J_{in}} \left( e^{-\lambda_k (T-j \cdot \tau_{net} - \tau_{in})} - e^{-\lambda_k (T-j \cdot \tau_{net})} \right)}^{\text{Partial ex-core}} \\
 & + \underbrace{e^{-\lambda_k t} \sum_{j=J_{in}+1}^J \left( 1 - e^{-\lambda_k (T-j \cdot \tau_{net})} \right)}_{\text{Partial in-core}}
 \end{aligned} \tag{16}$$

where

$T$  = the total irradiation time

$$J_{in} = \left\lfloor \frac{T - \tau_{in}}{\tau_{net}} \right\rfloor$$

= the number of recirculated in-core irradiations [#]

# Long cycle time elements have a negligible impact



Group half-lives:

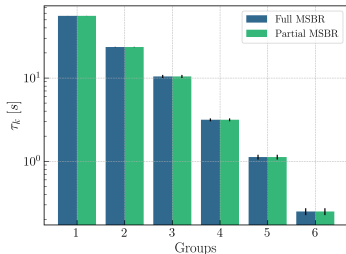


Figure 12:  $\tau_k$  for each DNP group  $k$  after a thermal saturation irradiation of  $^{235}\text{U}$  using the 0D scaled model.

Group yields:

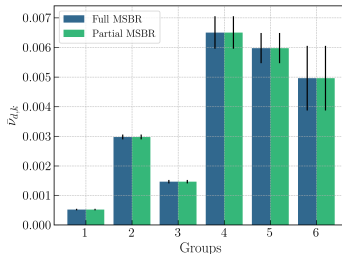


Figure 13:  $\bar{v}_{d,k}$  for each DNP group  $k$  after a thermal saturation irradiation of  $^{235}\text{U}$  using the 0D scaled model.

**The differences induced by exclusion of removal rates with cycle times longer than 20 seconds are minimal and well within uncertainty bounds.**

# Removal rates have a small impact in the 0D scaled model



Group half-lives:

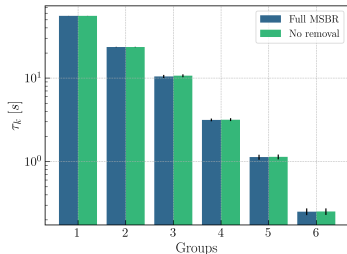


Figure 14:  $\tau_k$  for each DNP group  $k$  after a thermal saturation irradiation of  $^{235}\text{U}$  using the 0D scaled model.

Group yields:

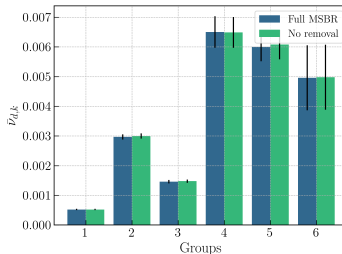


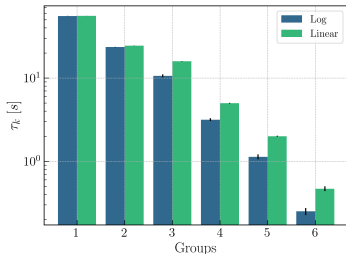
Figure 15:  $\bar{v}_{d,k}$  for each DNP group  $k$  after a thermal saturation irradiation of  $^{235}\text{U}$  using the 0D scaled model.

**Neglecting chemical removal rates generally leads to larger yields, though all differences are within uncertainty bounds.**

# Logarithmic spacing of time nodes give more accurate results

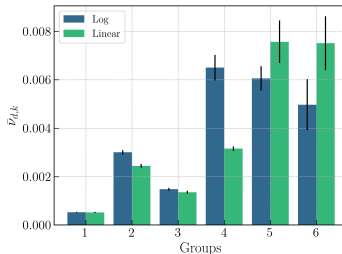


Group half-lives:



**Figure 16:**  $\tau_k$  for each DNP group  $k$  after a thermal saturation irradiation of  $^{235}\text{U}$  using the 0D scaled model.

Group yields:



**Figure 17:**  $\bar{\nu}_{d,k}$  for each DNP group  $k$  after a thermal saturation irradiation of  $^{235}\text{U}$  using the 0D scaled model.

**Linear spacing has no data between zero and two seconds, and it places greater weight on longer-lived groups due to the data distribution.**

# More time nodes give slightly more accurate results



Group half-lives:

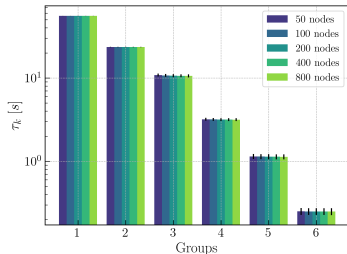


Figure 18:  $\tau_k$  for each DNP group  $k$  after a thermal saturation irradiation of  $^{235}\text{U}$  using the 0D scaled model.

Group yields:

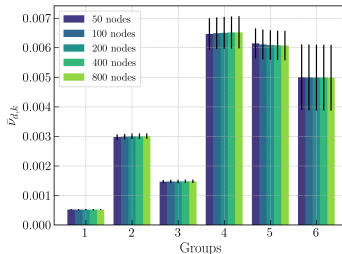


Figure 19:  $\bar{\nu}_{d,k}$  for each DNP group  $k$  after a thermal saturation irradiation of  $^{235}\text{U}$  using the 0D scaled model.

**Increasing the number of time nodes increases the accuracy of the DNP group parameters, with differences within uncertainty bounds.**

# Longer measurement times give more accurate results



## Group half-lives:

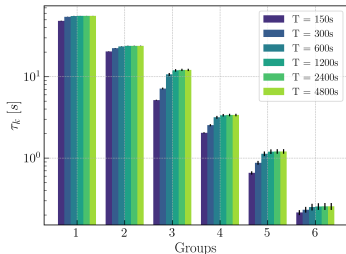


Figure 20:  $\tau_k$  for each DNP group  $k$  after a thermal saturation irradiation of  $^{235}\text{U}$  using the 0D scaled model.

## Group yields:

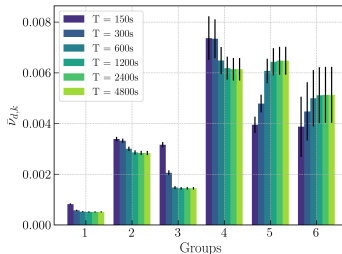


Figure 21:  $\bar{v}_{d,k}$  for each DNP group  $k$  after a thermal saturation irradiation of  $^{235}\text{U}$  using the 0D scaled model.

**Longer post-irradiation time results in longer-lived groups with smaller yields in groups one through four and larger yields in groups five and six.**

# The density of time nodes does not drive the group parameters



## Group half-lives:

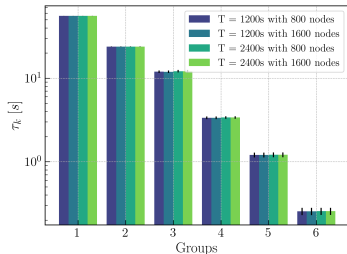


Figure 22:  $\tau_k$  for each DNP group  $k$  after a thermal saturation irradiation of  $^{235}\text{U}$  using the 0D scaled model.

## Group yields:

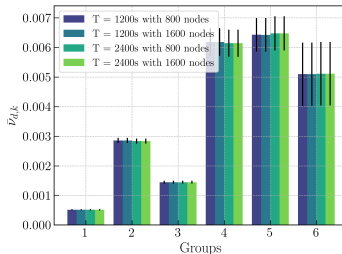


Figure 23:  $\bar{v}_{d,k}$  for each DNP group  $k$  after a thermal saturation irradiation of  $^{235}\text{U}$  using the 0D scaled model.

**The number of nodes has a small impact on the DNP group parameters, whereas the total decay time causes larger differences.**

# Safety-related transients may disrupt safe reactor operation



## Safety-related transients that will be modeled are:

- ULOHS
  - Loss of cooling in the reactor, potentially due to a heat exchanger failure
  - Modeled in Moltres by setting inlet temperature equal to outlet temperature
- ULOF
  - Flow rate decreases to zero due to a pump failure
  - Modeled in Moltres by changing flow field over time to match MSRE pump shut-down
- UTOP
  - Reactivity insertion event, which could be caused by poison removal, fissile addition, or temperature change
  - Modeled in Moltres by having a step-decrease in the removal cross section, representing control rod withdrawal
- FSOC
  - Fuel salt temperature decreases, potentially caused by increased heat removal rate
  - Modeled in Moltres by reducing inlet temperature by 25%

## Sabotage may be used as a diversion

This framework assumes a hostile actor and an insider with the objective to disrupt operations [32].

### Sabotage-initiated transients that will be modeled are:

- All safety-related transients as single events
- Combined UTOP and ULOHS
  - This should cause a temperature increase that propagates due to lack of heat sink; reactivity should then decrease due to temperature increase
- Combined UTOP and ULOF
  - This should cause a temperature increase with a peak maximum temperature due to the stationary fuel salt; reactivity should then decrease due to temperature increase
- Combined UTOP and FSOC
  - This should cause a temperature increase that is counteracted by the overcooled inlet fuel salt, leading to the largest reactivity change; temperature is not expected to change dramatically